

Why Sidewalk Sidekick?

43 million
people worldwide are blind

[1]

295 million
people worldwide live with
moderate-to-severe
visual impairment

[1]

\$30,000 - \$60,000*
average cost of
training & placing a guide dog
*before insurance and
government subsidies

[2]

Value Proposition

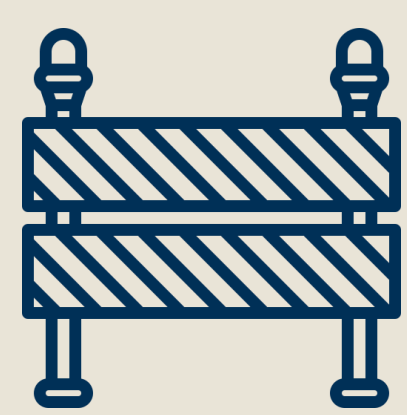
Blind & visually impaired individuals experience difficulties navigating urban terrain due to:



Scooters and trash bags
left in the middle of the sidewalk



Tree roots and uneven pavement that may not be detected by a white cane



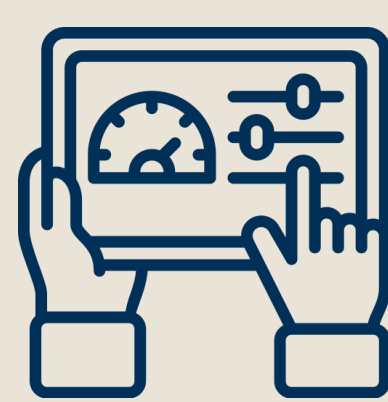
Inconsistent infrastructure, like dead-end crosswalks or construction zones

This raises the need for an integrated platform capable of **replicating the feel of a sighted guide**.



Environmental contextualization and obstacle detection

Customizable preferences for communication and control



References:

[1] <https://www.orbis.org/en/news/2021/new-global-blindness-data#:~:text=43%20million%20people%20live%20with%20blindness%20of,with%20moderate%2Dto%2Dsevere%20visual%20impairment%20from%20avoidable%20causes.>

[2] <https://www.guidedogsofamerica.org/frequently-asked-questions/>



Sidewalk Sidekick

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Aerospace Robotics Lab

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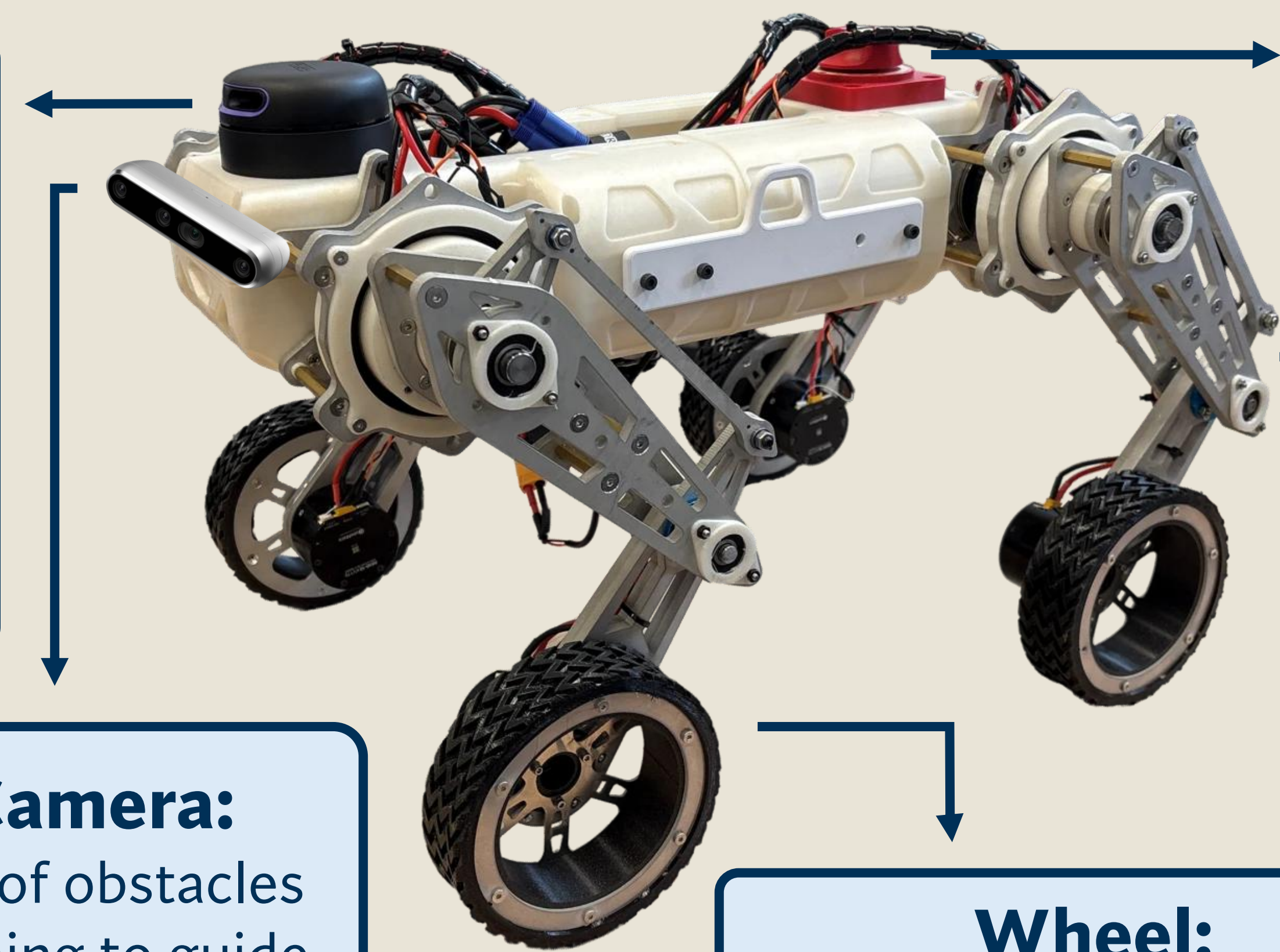
A Mobile Robotic Platform for Guidance

LIDAR:

provides increased FOV for waist-level obstacles and allows for object detection independent of lighting conditions.

RGB Depth Camera:

enables perception of obstacles and local path planning to guide users and provide environmental context.



On & Off

Leg:

enables locomotion over uneven urban terrain in tandem with user.

Wheel:

enables efficient traversal over smooth terrain and provides more reliable proprioceptive cues.

Key Performance Metrics



Maintains stable operation on terrain with local surface height variations up to **38mm**



Achieves minimum turning radius of **0.6m**



Supports ambulatory speeds up to **1.24 m/s**



Ingress Protection (IP) rating **IP54**
*not yet implemented



Detects obstacles within **0 – 5m range** and from **ground level to 2m**
*partially implemented



Both handle and robotic platform last for **multiple round trips** on a single charge



Voice command recognition with up to **73.4 dBA background noise**



Able to traverse vertical discontinuities between **100 – 240mm**
*partially implemented



Device weight is less than **15kg**



Final product costs less than the initial expense of a guide dog
*not yet implemented

User Testing

We were able to test our MVP with a blind Georgia Tech student, getting feedback on her:

- Preference for a **rigid handle vs. a flexible (pull-strap) handle**
- **Physical comfort** while using the device, especially with regards to joint strain
- Ability to **understand proprioceptive cues**



B Ergonomic Handle



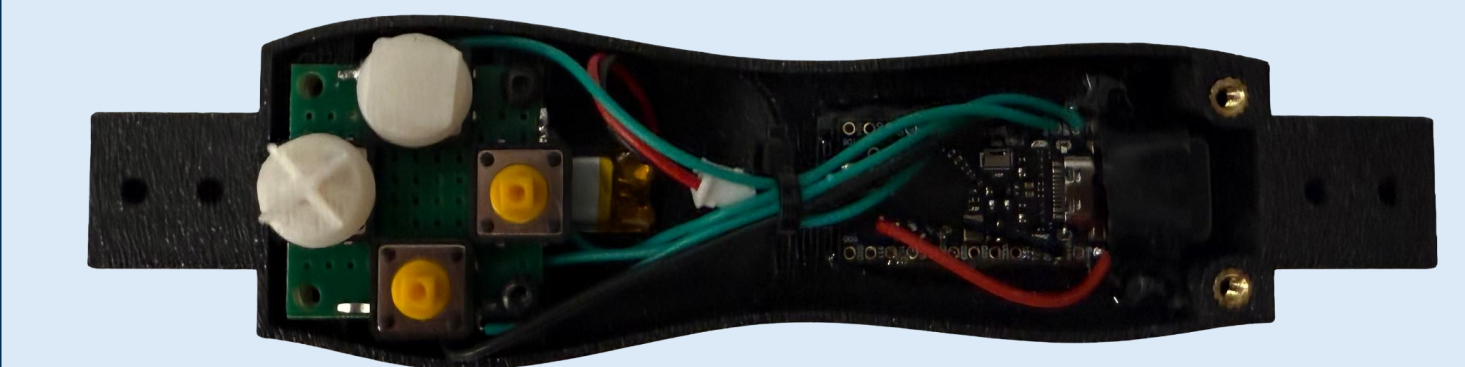
Bluetooth Headset:

allows for two-way communication between user and device.



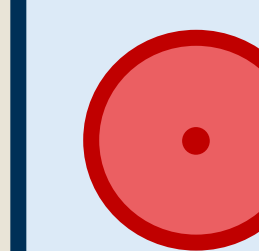
Integrated Electronics:

lightweight and tightly integrated handle with microcontroller, buttons, and USB-C charging port.

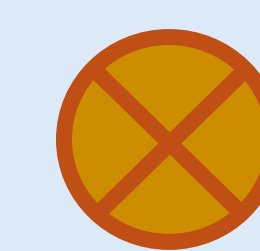


Manual Control Buttons:

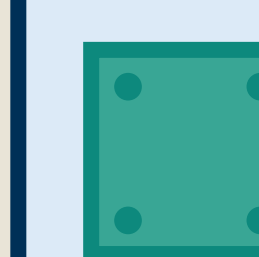
- **Tactile distinction:** buttons incorporate tactile bumps and raised shapes for easy identification.
- **Color contrast:** button colors are suited to a colorblind-friendly palette for those with lower levels of visual impairment.



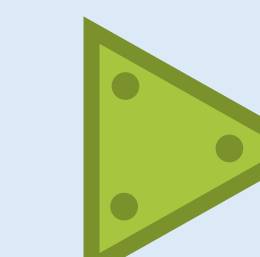
Stop route



Describe surroundings



Audio command



Start route



Next Steps

- GPS integration to improve path accuracy & safety
- Integration of user preference module into software
- Improving YOLO model and LIDAR integration for stronger environmental contextualization